THE SHEAFIFICATION OF A PRESHEAF OF SETS IS A SHEAF

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Pi(F) is a sheaf: compatible pointwise sections over a covering glue uniquely. invoked at proof lines 1769-1771. Provides the pointwise section s over U restricting to the s_i.
- Property (*) for F# sections: around each point the representing families come from a genuine section of F on a smaller open. invoked at proof lines 1773-1777. The defining property verified for the glued section s.

Source reference. Original remark 006V: sources/stacks-sheaves/pinned-original-source.tex, source lines 771–771. Cited at proof lines 1760-1763 as an analogy only; the actual verification is carried out inside the counted span and does not use the example.

Lemma 0.2. *The presheaf $\mathcal{F}^\#$ is a sheaf.*

Proof. It is probably better for the reader to find their own explanation of this than to read the proof here. In fact the lemma is true for the same reason as why the presheaf of continuous function is a sheaf, see Example 006V (and this analogy can be made precise using the "espace étalé").

Anyway, let $U = \bigcup U_i$ be an open covering. Suppose that $s_i = (s_{i,u})_{u \in U_i} \in \mathcal{F}^\#(U_i)$ such that s_i and s_j agree over $U_i \cap U_j$. Because $\Pi(\mathcal{F})$ is a sheaf, we find an element $s = (s_u)_{u \in U}$ in $\prod_{u \in U} \mathcal{F}_u$ restricting to s_i on U_i. We have to check property $(*)$. Pick $u \in U$. Then $u \in U_i$ for some i. Hence by $(*)$ for s_i, there exists a V open, $u \in V \subset U_i$ and a $\sigma \in \mathcal{F}(V)$ such that $s_{i,v} = (V, \sigma)$ in $\mathcal{F}_v$ for all $v \in V$. Since $s_{i,v} = s_v$ we get $(*)$ for s. □

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